



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

CHEMISTRY

PAPER 1 Multiple Choice

6031/1

NOVEMBER 2022 SESSION

1 hour

Additional materials:

Data Booklet

Mathematical tables and/or Electronic calculator

Multiple Choice answer sheet

Soft pencil (type B or HB is recommended)

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, Centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A, B, C** and **D**. Choose the one you consider correct and record your choice in soft pencil on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score **one** mark. A mark will **not** be deducted for a wrong answer. Any rough working should be done in this booklet.

This question paper consists of 13 printed pages and 3 blank pages.

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Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the one you consider to be correct.

- 1 The table shows physical properties of four substances.

Which substance, **A**, **B**, **C** or **D**, could be calcium chloride?

	electrical conductivity of solid	electrical conductivity of liquid	solubility in water	melting point/°C
A	poor	poor	insoluble	-119
B	poor	good	insoluble	-78
C	good	good	soluble	213
D	poor	good	soluble	993

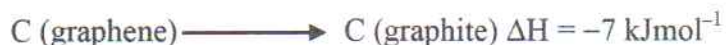
- 2 What are the bond angles in PCl_5 ?

- A** 72° and 90°
B 72° and 107°
C 90° and 109.5°
D 90° and 120°

- 3 Which is the correct order of increasing lattice energy?

- A** $\text{LiF} < \text{MgO} < \text{RbCl}$
B $\text{RbCl} < \text{LiF} < \text{MgO}$
C $\text{MgO} < \text{LiF} < \text{RbCl}$
D $\text{RbCl} < \text{MgO} < \text{LiF}$

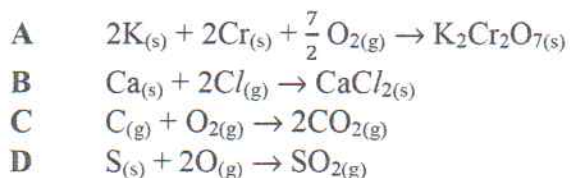
- 4 Graphene undergoes the chemical change shown:



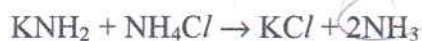
Which statement explains why the reaction is exothermic?

- A** the formation of graphite from graphene is endothermic
B the change from graphene to graphite has high activation energy
C the single layer in graphene is more stable than many layers in graphite
D graphene has only strong covalent bonds whereas graphite has both covalent bonds and Van der Waals forces

- 5 Which equation correctly describes the standard enthalpy change of formation of the product?



- 6 The following reaction takes place in a suitable solvent.



Why should the reaction be classified as a Bronsted-Lowry acid-base reaction?

- A ammonium ion is a proton acceptor
 B ammonium ion is a proton donor
 C ammonium ion is an electron-pair donor
 D ammonium ion is an electron pair acceptor

- 7 The acid dissociation constant for a weak monobasic acid, HA, is $1.2 \times 10^{-4} \text{ mol dm}^{-3}$.

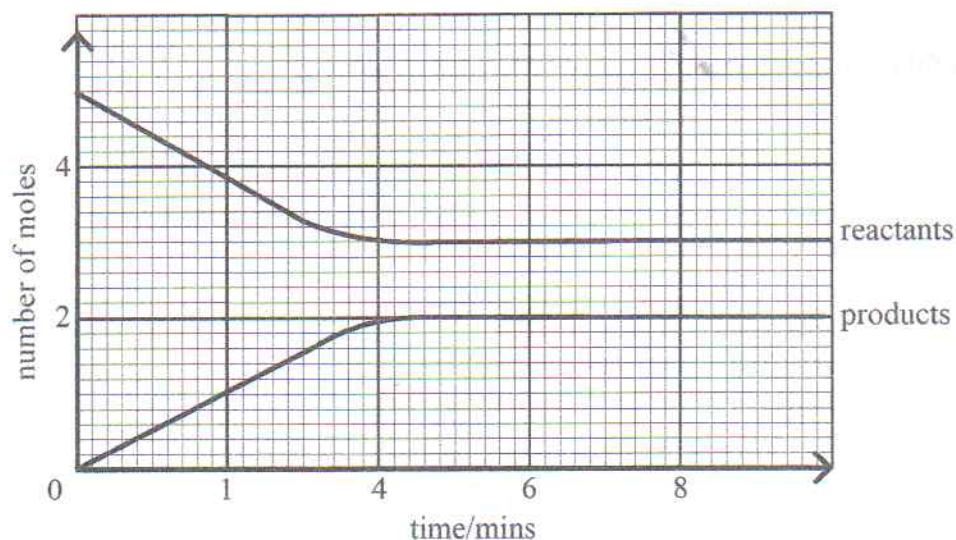
Which expression gives the concentration of hydrogen ions in a 0.05 mol dm^{-3} of the weak acid?

- A $(1.2 \times 10^{-4})^2 \text{ mol dm}^{-3}$
 B $\left(\frac{1.2 \times 10^{-4}}{0.05}\right)^2 \text{ mol dm}^{-3}$
 C $\left(\frac{1.20 \times 10^{-4} \times 0.05}{0.05}\right)^{\frac{1}{2}} \text{ mol dm}^{-3}$
 D $\left(\frac{1.20 \times 10^{-4}}{0.05}\right)^{\frac{1}{2}} \text{ mol dm}^{-3}$

- 8 Compound Z decomposes according to the equation:



In an experiment 5 moles of Z were decomposed under certain conditions in a closed container and the results obtained are shown in the graph.



Which statement explains the shape of the graph after four minutes?

- A the reaction stopped
 - B compound Z was used up
 - C the concentration of Z and X had become equal
 - D the rate of forward and backward reactions were equal
- 9 A mass of 3.47 g of a carbonate, XCO_3 , were strongly heated and gave 0.564 dm^3 of a gaseous product at r.t.p.

What is the A_r of X?

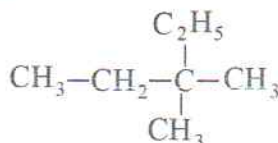
- A 23.0
- B 40.1
- C 65.4
- D 87.7

- 10 When 4 g of a group (II) nitrate, $M(\text{NO}_3)_2$ was decomposed, 1.37 g of the oxide was produced.

What is the identity of M?

- A beryllium
B magnesium
C calcium
D strontium
- 11 Which oxide forms the most alkaline solution?
- A MgO
B CaO
C SrO
D BaO
- 12 Which oxide of a Group (IV) element will react with hydrochloric acid?
- A CO
B CO_2
C SiO_2
D PbO
- 13 Which equation correctly represents the reaction between chlorine and hot concentrated aqueous sodium hydroxide?
- A $\text{Cl}_{(\text{aq})} + 4\text{OH}^-_{(\text{aq})} \rightarrow 2\text{Cl}^-_{(\text{aq})} + 2\text{H}_2\text{O}_{(\text{l})}$
B $\text{Cl}_{2(\text{aq})} + 2\text{OH}^-_{(\text{aq})} \rightarrow \text{Cl}^-_{(\text{aq})} + \text{ClO}^-_{(\text{aq})} + \text{H}_2\text{O}_{(\text{l})}$
C $3\text{Cl}_{2(\text{aq})} + 6\text{OH}^-_{(\text{aq})} \rightarrow 5\text{Cl}^-_{(\text{aq})} + \text{ClO}_{3(\text{aq})} + 3\text{H}_2\text{O}_{(\text{l})}$
D $4\text{Cl}_{2(\text{aq})} + 8\text{OH}^-_{(\text{aq})} \rightarrow 6\text{Cl}^-_{(\text{aq})} + \text{ClO}^-_{(\text{aq})} + \text{Cl}^-_{3(\text{aq})} + 4\text{H}_2\text{O}_{(\text{l})}$
- 14 Which substance is **not** obtained by the electrolysis of brine?
- A chlorine
B hydrogen
C oxygen
D sodium chlorate

- 15 Reactions of nitrogen with oxygen require very high temperatures due to the
- A half filled 2 p orbitals in nitrogen which makes it stable.
 - B electron-electron repulsion in one of the 2p orbitals of oxygen.
 - C strength of the bond in O₂ gas.
 - D strength of the bond in N₂ gas.
- 16 How many structural isomers does the alkane C₅H₁₂ have?
- A 1
 - B 2
 - C 3
 - D 4
- 17 What is the minimum number of carbon atoms must an alkane have to exhibit optical isomerism?
- A 1
 - B 5
 - C 6
 - D 7
- 18 An alkane has the structure shown.



What is the name of the alkane?

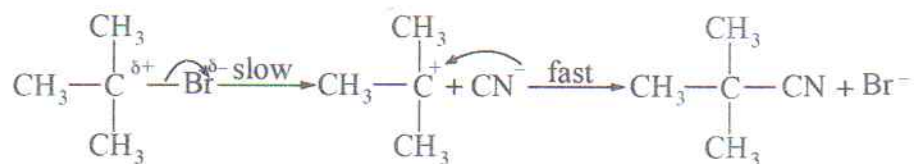
- A 3,3-dimethylpentane
- B 2,2-dimethylpentane
- C 2-ethyl-2-methylbutane
- D 2-methyl-2-ethylbutane

- 19 The diagram shows the conversion of a hydrocarbon to a chlorohydrocarbon.



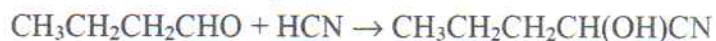
Which reagent and condition is needed to effect the conversion?

- A Cl_2 in FeCl_3
 - B Cl_2 , uv light
 - C Cl_2 , inert solvent
 - D SOCl_2 , reflux
- 20 The halogenoalkane $(\text{CH}_3)_3\text{Br}$ reacts with $\text{NaCN}_{(\text{aq})}$ through the mechanism shown.



The type of mechanism shown is nucleophilic

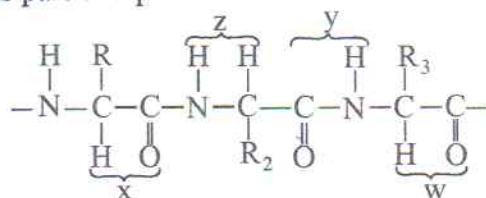
- A addition by SN_1 .
 - B addition by SN_2 .
 - C substitution by SN_2 .
 - D substitution by SN_1 .
- 21 Butanal can react with hydrogen cyanide as shown.



What type of chemical reaction takes place?

- A electrophilic addition
- B free radical substitution
- C electrophilic substitution
- D nucleophilic addition

- 22 The structure shown is part of a protein molecule.

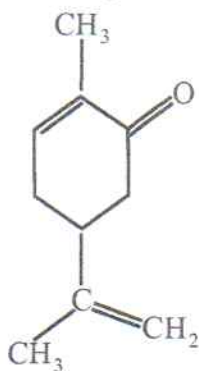


Which part, W, X, Y or Z, is hydrolysed by dilute hydrochloric acid?

- A W
 B X
 C Y
 D Z
- 23 Which sequence correctly shows the increasing order of basicity of the compounds shown?

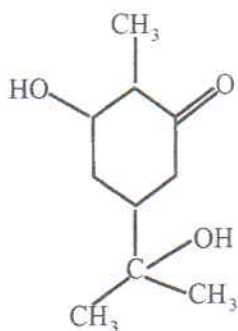
	least basic		most basic
A	$(\text{CH}_3)_2\text{NH}$	$\text{C}_6\text{H}_5\text{NH}_2$	NH_3
B	$\text{C}_6\text{H}_5\text{NH}_2$	NH_3	$(\text{CH}_3)_2\text{NH}$
C	$(\text{CH}_3)_2\text{NH}$	NH_3	$\text{C}_6\text{H}_5\text{NH}_2$
D	NH_3	$\text{C}_6\text{H}_5\text{NH}_2$	$(\text{CH}_3)_2\text{NH}$

- 24 The structural formula of an organic compound is shown.

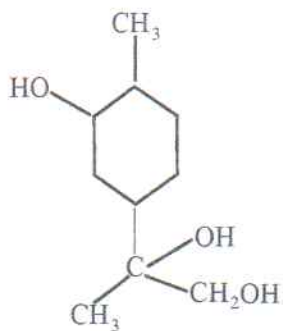


What is the formula of the organic product formed when the compound reacts with steam?

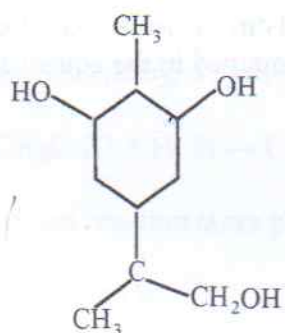
A



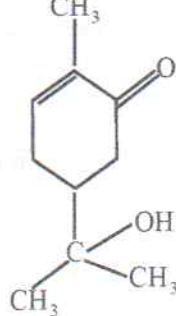
B



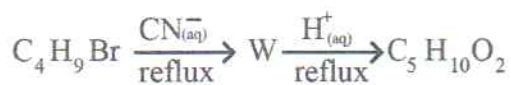
C



D



- 25 The compound C_4H_9Br undergoes a sequence of reactions as follows:



What could be the formula of W?

- A $CH_3CH_2CH_2CH_2CN$
 B $CH_3CH_2CH_2NH_2$
 C $CH_3CH_2CH_2CH_2CN_2$
 D $CH_3CH_2CH_2CH_2CH_2OH$

26 What is the challenge in using a radioactive material?

- A drawing out energy
- B fusion of particles
- C disposal of nuclear waste
- D ☒ handling and storage of the material

27 How is radioactive waste disposed?

- A In specific packaging criteria set out by the Atomic Energy Control Act.
- B ☒ In containers lined with yellow bags and labelled radioactive waste.
- C In containers lined with transparent bags and labelled radioactive waste.
- D In containers lined with black bags and labelled radioactive waste.

28 Which nanomaterial has different properties from its bulk form?

- A bacteria
- B ☒ viruses
- C graphene
- D gold

29 Methane is a major component of landfill gas.

Which statement about methane gas is correct?

- A It is a major contaminant of underground water supplies.
- B ☒ It may be harnessed and used to produce energy.
- C It must be pumped to the surface in leachate pipes.
- D It is produced when waste reacts with oxygen.

30 A solution of 10 g of an organic acid in 100 cm³ of water was shaken with 100 cm³ of ether. It was found that 6.0 g of the acid remained in the aqueous solution at equilibrium.

What is the partition coefficient?

- A 0.06
- B 0.40
- C 0.60
- D ☒ 0.67

Section B

For each of the questions in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct. (You may find it helpful to put a tick against the statement(s) which you consider to be correct).

The responses A to D should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 31 A sample of chlorine gas was analysed using mass spectrometry and gave three peaks with m/e values corresponding to 70, 72 and 74.

What is the ratio of the peak heights obtained respectively?

1. 9: 6: 1

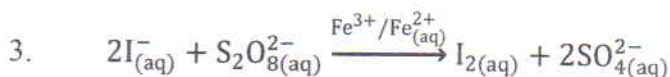
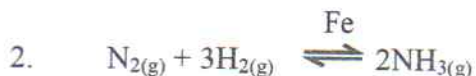
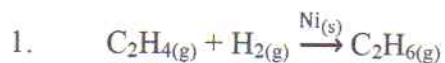
2. 27: 9: 6

3. 1: 1: 1

- 32 What determines the colour of a transition metal complex?

- the metal in the complex
- oxidation state of the metal in the complex
- type of the ligand

- 33 Which conversion involves heterogenous catalysis?

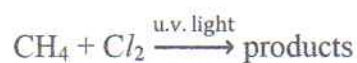


34 Which compound(s) react(s) with

- (i) sodium metal to release hydrogen,
- (ii) propanoylchloride to give hydrogen chloride.

- 1. C_6H_5OH
- 2. $(CH_3)_3C(OH)$
- 3. $CH_3CH_2CH_2OH$

35 Methane reacts with Cl_2 in the presence of u.v. light as shown by the equation:



The first stage in the mechanism does **not** involve fission of CH_4 to form $CH_3\bullet$ and $H\bullet$ radical because

- 1. methane is a transparent gas and **cannot** absorb u.v light.
- 2. the Cl-Cl bond energy is lower than the C-H bond energy in methane.
- 3. $CH_3\bullet$ can only be formed in the propagation step.

36 Two gaseous products are formed when decane, $C_{10}H_{22}$, is passed over an alumina bed at $450 - 500^\circ C$.

Which pair(s) correctly represent the possible products of the reaction?

- 1. C_8H_{18} and C_2H_6
- 2. C_3H_8 and C_7H_{16}
- 3. C_6H_{14} and C_4H_{10}

37 Nanotubes have

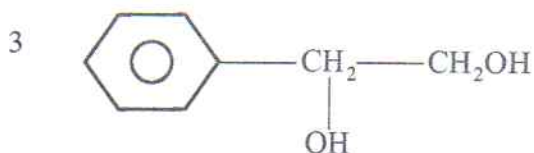
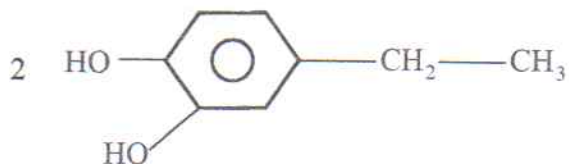
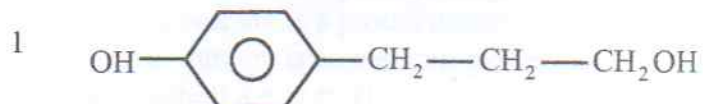
- 1. high electrical conductivity
- 2. high melting point
- 3. high tensile strength

38 Which statement(s) correctly describe(s) the effect of nitrogen compounds on the environment?

1. increases the biological Oxygen Demand (BOD) of water bodies
2. reduces the body's ability to absorb oxygen in the lungs
3. are major components in the formation of acid rain

39 An organic compound, X, produces an aqueous solution with pH less than 7 and gives 1 mole of H_2 gas per mole of X when reacted with excess sodium.

Which compound(s) could be X?



40 Which procedure(s) can reduce vanadium (V) to vanadium (IV)?

1. addition of iron (II) salts
2. bubbling sulphuric dioxide
3. addition of zinc metal